

A Practical Guide to Machine translation

TOPIC -- MACHINE TRANSLATION

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Word-sense disambiguation concerns finding a suitable translation when a word can have more than one meaning. The problem was first raised in the 1950s by Yehoshua Bar-Hillel. He pointed out that without a "universal encyclopedia", a machine would never be able to distinguish between the two meanings of a word. Today there are numerous approaches designed to overcome this problem. They can be approximately divided into "shallow" approaches and "deep" approaches. Shallow approaches assume no knowledge of the text. They simply apply statistical methods to the words surrounding the ambiguous word. Deep approaches presume a comprehensive knowledge of the word. So far, shallow approaches have been more successful. Claude Piron, a long-time translator for the United Nations and the World Health Organization, wrote that machine translation, at its best, automates the easier part of a translator's job; the harder and more time-consuming part usually involves doing extensive research to resolve ambiguities in the source text, which the grammatical and lexical exigencies of the target language require to be resolved:

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There are many factors that affect how machine translation systems are evaluated. These factors include the intended use of the translation, the nature of the machine translation software, and the nature of the translation process. Different programs may work well for different purposes. For example, statistical machine translation (SMT) typically outperforms example-based machine translation (EBMT), but researchers found that when evaluating English to French translation, EBMT performs better. The same concept applies for technical documents, which can be more easily translated by SMT because of their formal language. In certain applications, however, e.g., product descriptions written in a controlled language, a dictionary-based machine-translation system has produced satisfactory translations that require no human intervention save for quality inspection. There are various means for evaluating the output quality of machine translation systems. The oldest is the use of human judges to assess a translation's quality. Even though human evaluation is time-consuming, it is still the most reliable method to compare different systems such as rule-based and statistical systems. Automated means of evaluation include BLEU, NIST, METEOR, and LEPOR. Relying exclusively on unedited machine translation

ignores the fact that communication in human language is context-embedded and that it takes a person to comprehend the context of the original text with a reasonable degree of probability. It is certainly true that even purely human-generated translations are prone to error. Therefore, to ensure that a machine-generated translation will be useful to a human being and that publishable-quality translation is achieved, such translations must be reviewed and edited by a human. The late Claude Piron wrote that machine translation, at its best, automates the easier part of a translator's job; the harder and more time-consuming part usually involves doing extensive research to resolve ambiguities in the source text, which the grammatical and lexical exigencies of the target language require to be resolved. Such research is a necessary prelude to the pre-editing necessary in order to provide input for machine-translation software such that the output will not be meaningless. In addition to disambiguation problems, decreased accuracy can occur due to varying levels of training data for machine translating programs. Both example-based and statistical machine translation rely on a vast array of real example sentences as a base for translation, and when too many or too few sentences are analyzed accuracy is jeopardized. Researchers found that when a program is trained on 203,529 sentence pairings, accuracy actually decreases. The optimal level of training data seems to be just over 100,000 sentences, possibly because as training data increases, the number of possible sentences increases, making it harder to find an exact translation match. Flaws in machine translation have been noted for their entertainment value. Two videos uploaded to YouTube in April 2017 involve two Japanese hiragana characters ■■ (e and gu) being repeatedly pasted into Google Translate, with the resulting translations quickly degrading into nonsensical phrases such as "DECEARING EGG" and "Deep-sea squeeze trees", which are then read in increasingly absurd voices; the full-length version of the video currently has 7.1 million views as of August 2025.

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Transfer-based machine translation was similar to interlingual machine translation in that it created a translation from an intermediate representation that simulated the meaning of the original sentence. Unlike interlingual MT, it depended partially on the language pair involved in the translation.

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1975–1980s SYSTRAN, which "pioneered the field under contracts from the U.S. government" in the 1960s, was used by Xerox to translate technical manuals (1978). Beginning in the late 1980s, as computational power increased

and became less expensive, more interest was shown in statistical models for machine translation. MT became more popular after the advent of computers. SYSTRAN's first implementation system was implemented in 1988 by the online service of the French Postal Service called Minitel. Various computer based translation companies were also launched, including Trados (1984), which was the first to develop and market Translation Memory technology (1989), though this is not the same as MT. The first commercial MT system for Russian / English / German-Ukrainian was developed at Kharkov State University (1991).

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Copyright Only works that are original are subject to copyright protection, so some scholars claim that machine translation results are not entitled to copyright protection because MT does not involve creativity. The copyright at issue is for a derivative work; the author of the original work in the original language does not lose his rights when a work is translated: a translator must have permission to publish a translation.

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Law Legal language poses a significant challenge to machine translation tools due to its precise nature and atypical use of normal words. For this reason, specialized algorithms have been developed for use in legal contexts. Due to the risk of mistranslations arising from machine translators, researchers recommend that machine translations should be reviewed by human translators for accuracy, and some courts prohibit its use in formal proceedings. The use of machine translation in law has raised concerns about translation errors and client confidentiality. Lawyers who use free translation tools such as Google Translate may accidentally violate client confidentiality by exposing private information to the providers of the translation tools. In addition, there have been arguments that consent for a police search that is obtained with machine translation is invalid, with different courts issuing different verdicts over whether or not these arguments are valid.